D Y PATIL AGRICULTURE & TECHNICAL UNIVERSITY TALSANDE

MCA SANDWICH

PROJECT NAME

VISUALIZING HOUSING MARKETS TRENDS: AN ANALYSIS OF SALE PRICES AND FEATURES USING TABLEAU

TEAM MEMBERS

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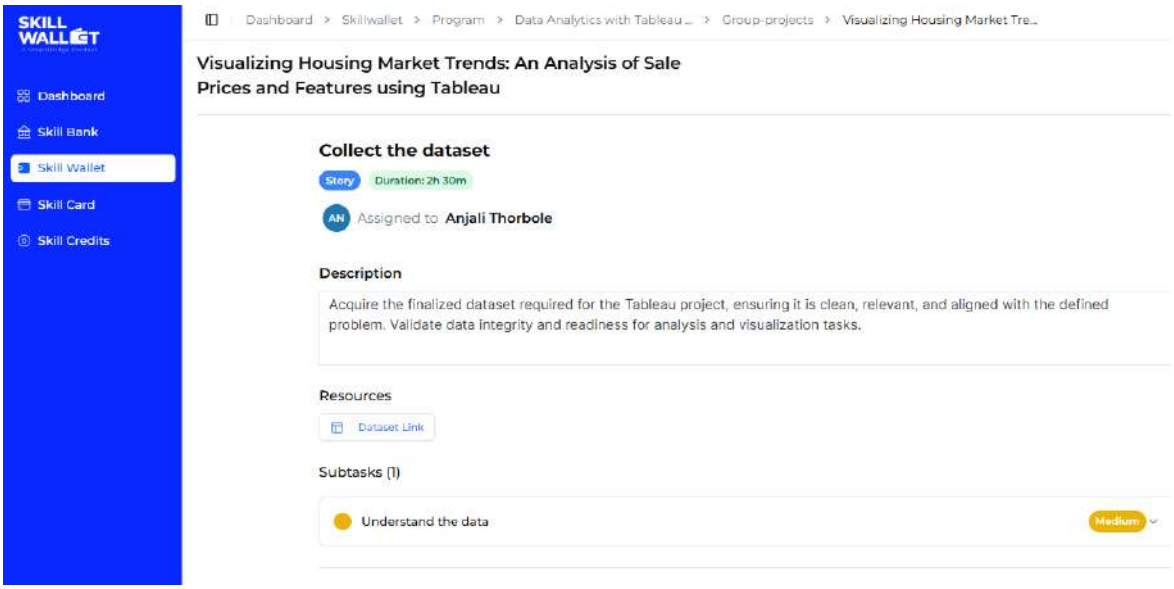
2) ANJALI THORBOLE
3) NEHA YADAV
4) PRANAV PATIL

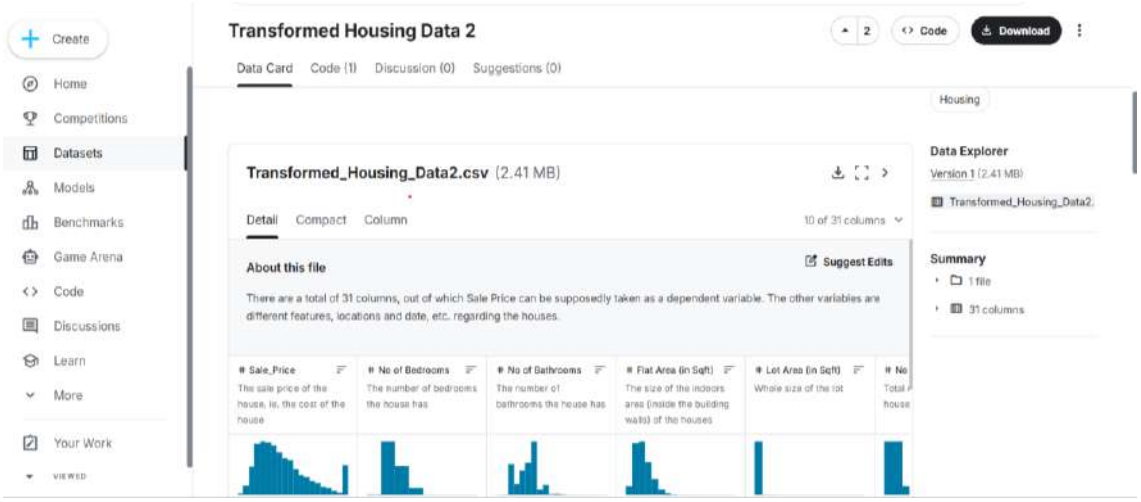
1) Data Collection & Extraction from Database:

❖ Collect the dataset:

1) Data is Collected from Dataset link Which is provided already so click on **Dataset Link**.



2) From the Kaggle Download the Dataset **Transformed_Housing_Data2.csv** by clicking on **Download** button,



❖ Loading the Dataset:

Step 1: Click on "Microsoft Excel" or "Text File"

On the Start Page, under Connect, you will see options like:

Microsoft Excel (for .xlsx)

Text File (for .csv)

Click the correct option.

Step 2: Select Your File

Browse your computer.

Select your dataset file.

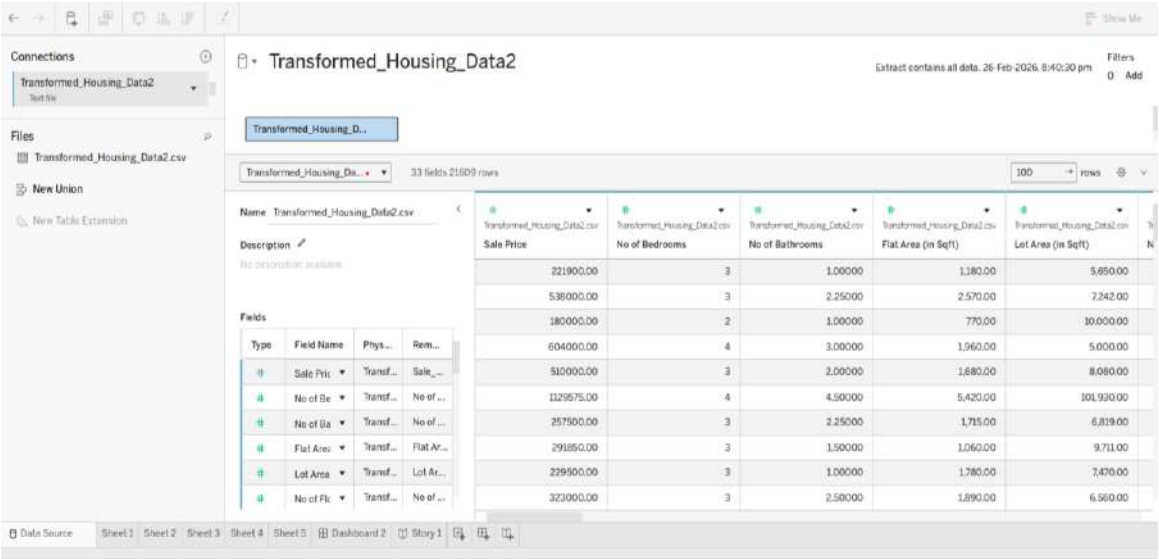
Click Open.

Step 3: Data Source Page

Your sheet names will appear on the left side.

Drag the sheet to the centre area.

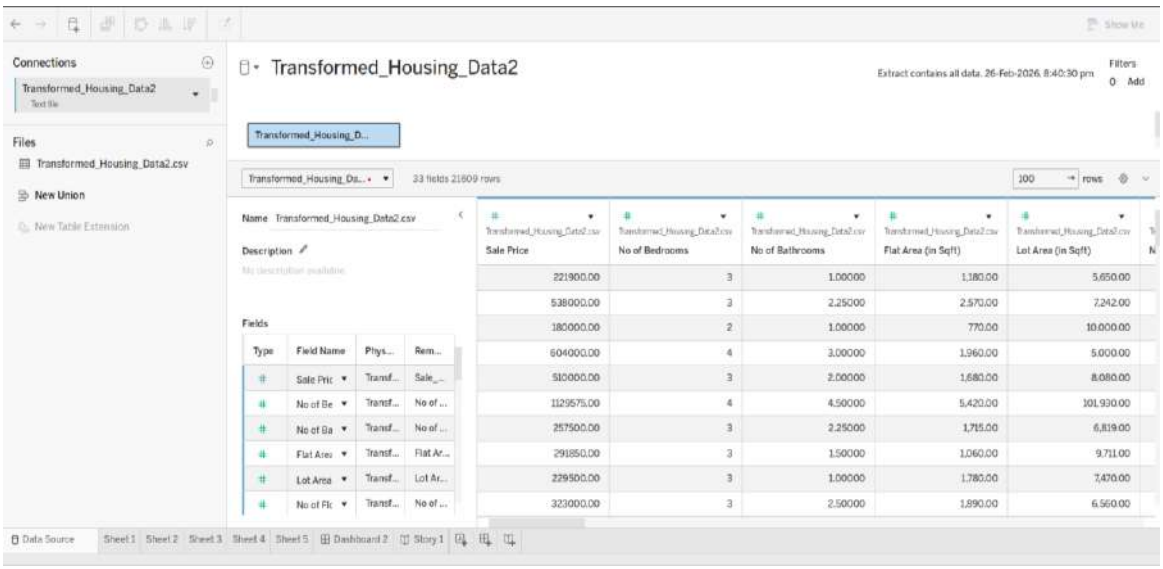
Tableau will show a preview of your data.



2)Data Preparation:

❖ Data Cleaning

Preparing the data for visualization involves cleaning the data to remove irrelevant or missing data, transforming the data into a format that can be easily visualized, exploring the data to identify patterns and trends, filtering the data to focus on specific subsets of data, preparing the data for visualization software, and ensuring the data is accurate and complete.



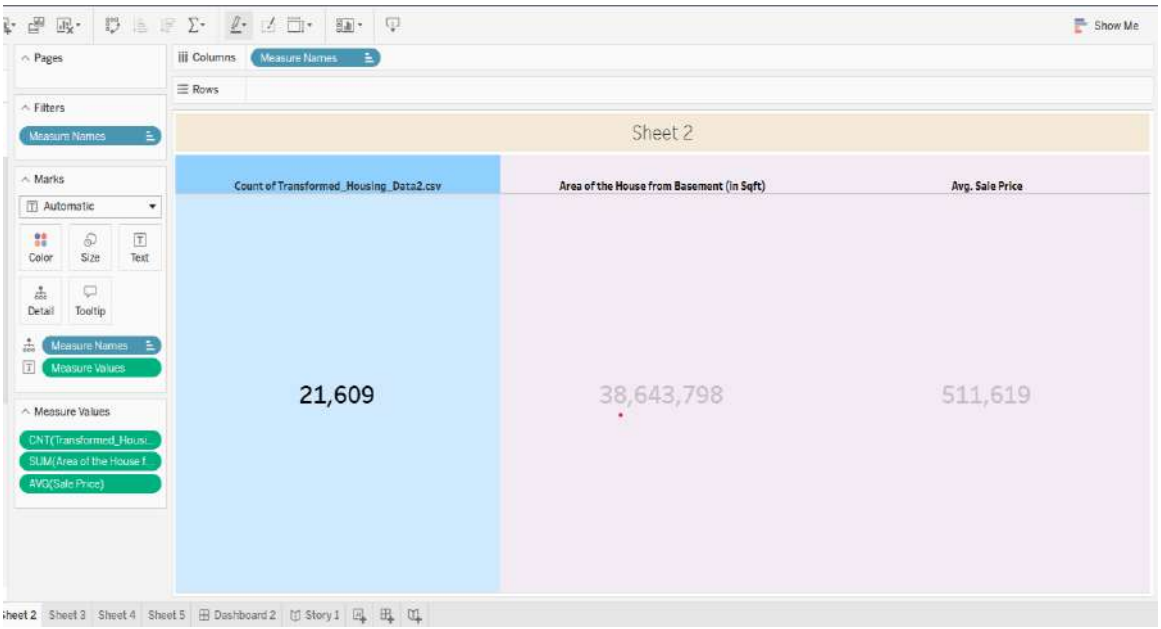
2) Data Visualization:

Data visualization is the process of creating graphical representations of data to help people understand and explore the information. The goal of data visualization is to make complex data sets more accessible, intuitive, and easier to interpret. By using visual elements such as charts, graphs, and maps, data visualizations can help people quickly identify patterns, trends, and outliers in the data.

❖ No of Unique Visualizations:

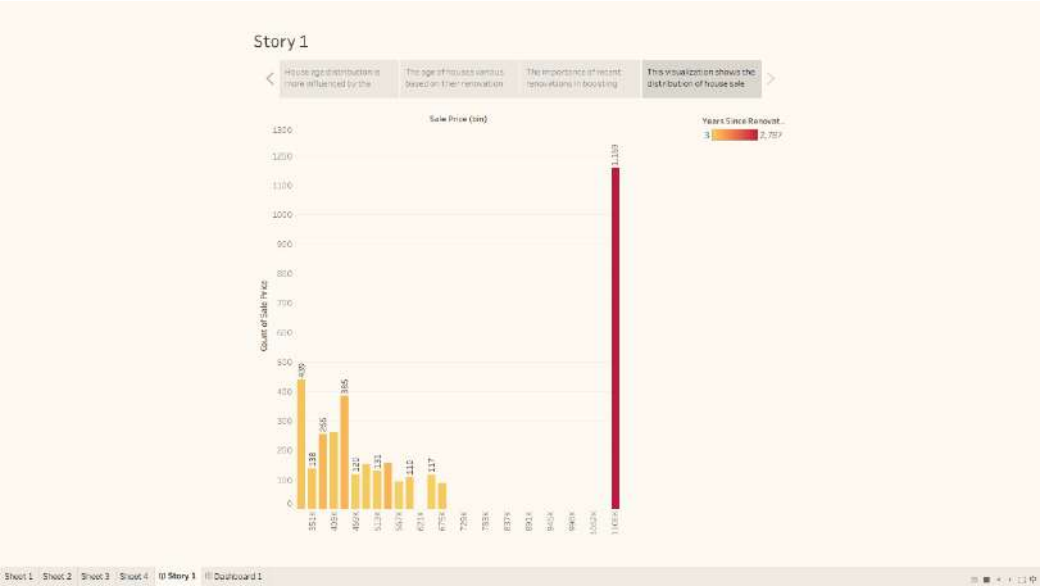
1) Count of Transformed_Housing_Data2.cvs

a) Drag and Drop the Measure Name and Measure Values.
b) Add only-
   - Count of Transformed_Housing_Data2.cvs
   - Area of the House from Basement (in Sqft)
   - Avg.SalePrice
c) Drag and Drop Measure Name at Column section.



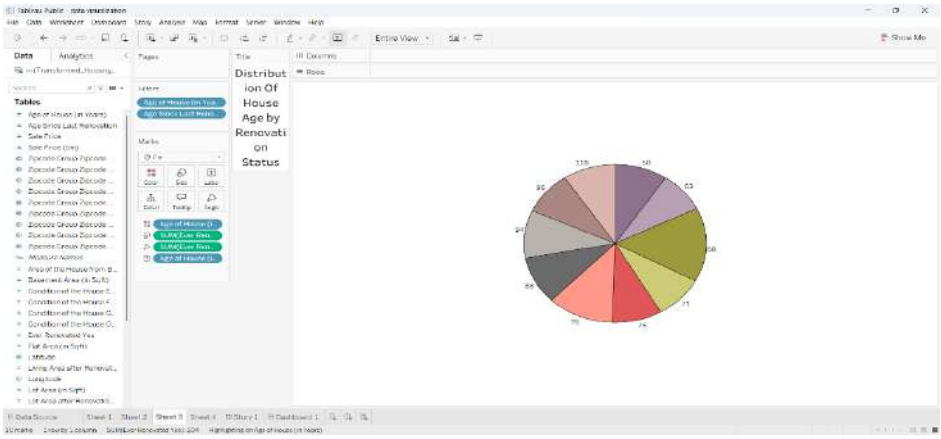
2)Total Sales by Years Since Renovation

a) Convert the Sales Price into Dimension by right clicking.
b) Add Filters into Sales Price and Sales Price(bin) to display Top 25 records.
c) Drag and Drop Sales Price(bin) for grouping Sales Price into column section, and also CNT(Sales price) into rows.
d) Drag and Drop SUM(Year Since Renovation) into color and SUM(Sale Price) into Lable.

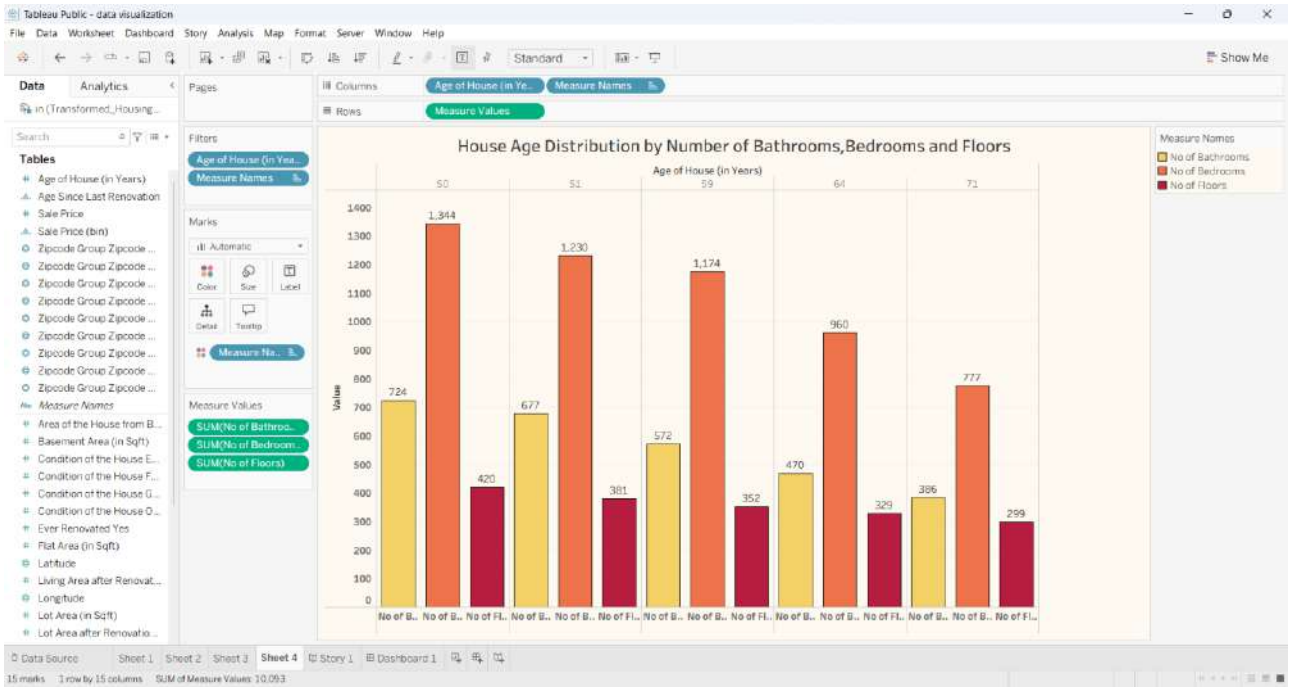


2) Distribution of House Age by Renovation Status

a) Convert Age of House(in Years) to Dimension.
b) Filter Age of House(in Years) into Top 10 records.
c) Add Age of House(in Years) into color.
d) Add SUM(Ever Renovated yes) into angle.
e) Add Sum(Ever Renovated yes) into size.
f) Add Age of House(in Years) in to label.



3) **House Age Distribution by Number of Bathrooms, Bedrooms and Floors**
   a) Drag and Drop Measure Name and Age of House(in Years).
   b) Filter Age of House(in Years) into Top 5 records.
   c) Drag and Drop Measure Values-
      - SUM(No of Bathroom)
      - SUM(No of Bedroom)
      - SUM(No of Floors)
   d) Drag and Drop Measure Names into Color.
   e) Add Age of House(in Year) and Measure Name into column section, and Measure Values into rows.



3) Dashboard

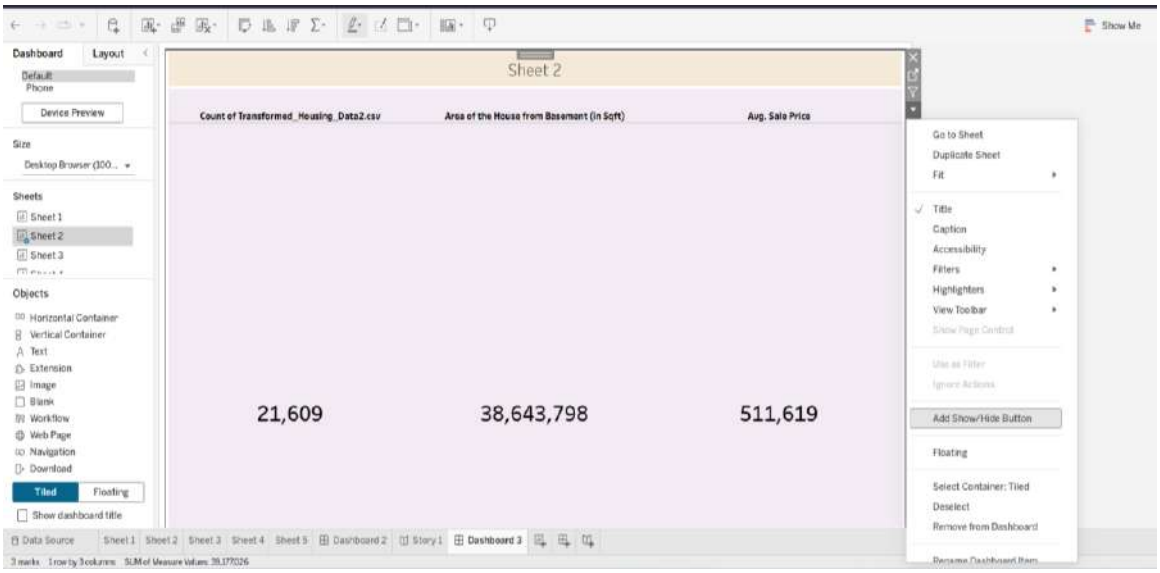
A dashboard is a graphical user interface (GUI) that displays information and data in an organized, easy-to-read format. Dashboards are often used to provide real-time monitoring and analysis of data and are typically designed for a specific purpose or use case. Dashboards can be used in a variety of settings, such as business, finance, manufacturing, healthcare, and many other industries. They can be used to track key performance indicators (KPIs), monitor performance metrics, and display data in the form of charts, graphs, and tables.

❖ Responsive and Design of Dashboard

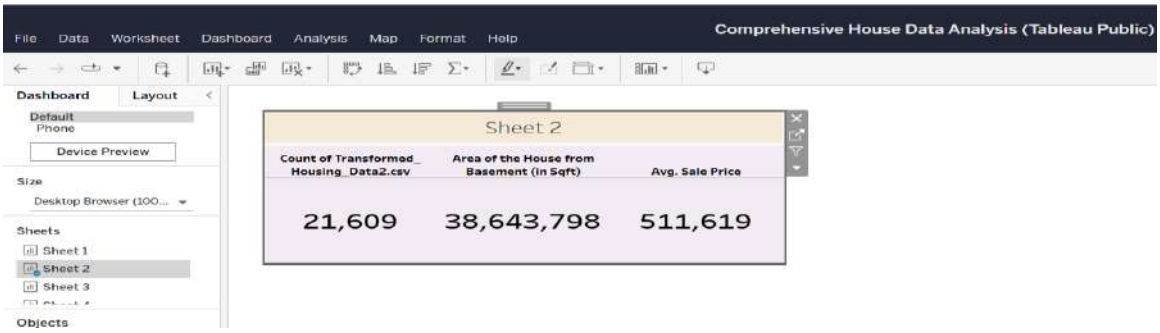
a) Click On **New Dashboard** option.

b) Drag and Drop Sheets from the data panel into centre white screen and in the right side of sheet there is Downward arrow so click on that and choose **Floating** option.

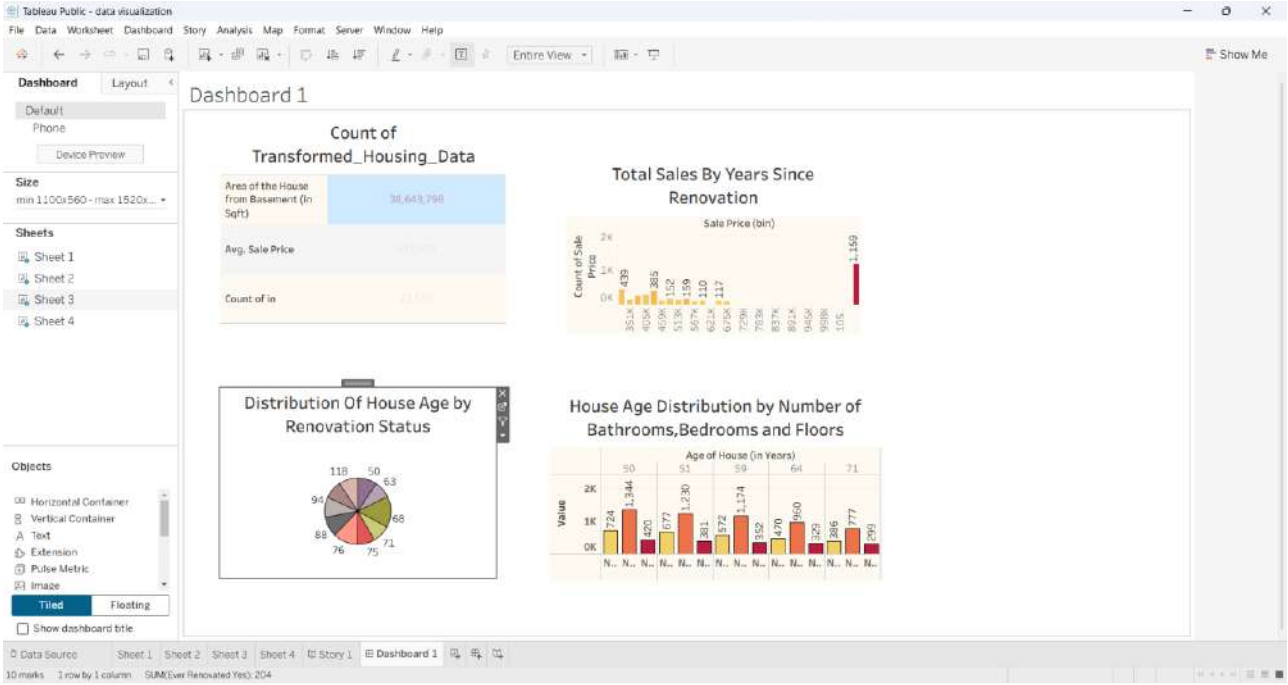
c)



d) After selecting Floating option it's looks like.



e) Drag and Drop the Required Sheets for dashboard there I added 4 sheets.



4) Storyboard

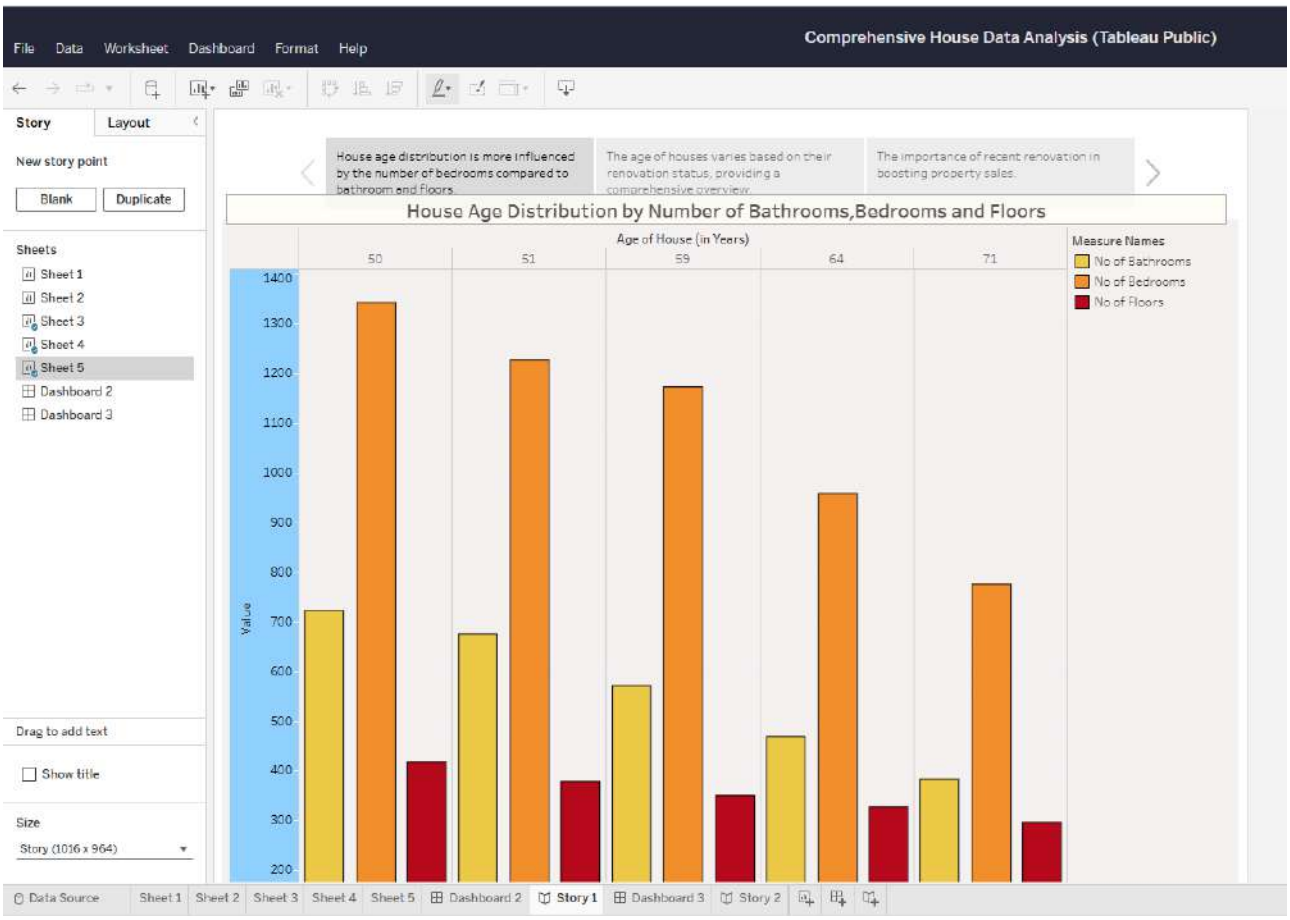
A storyboard is a visual representation of a sequence of events, typically used in multimedia projects such as films, animations, advertisements, or presentations. It consists of a series of drawings or images arranged in a sequence, often accompanied by annotations or descriptions, to outline the flow of the story or concept.

❖ Design a Storyboard

a) Click On **<u>New Story</u>** option.

b) Here I have to add 3 sheets into one story so click on **Blank** .

c) Drag and Drop Sheets from Story panel into white Screen and add caption for each sheet.


File Data Worksheet Dashboard Format Help
Comprehensive House Data Analysis (Tableau Public)
Story
Layout
New story point
Blank
Duplicate
Sheets
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Dashboard 2
Dashboard 3
House age distribution is more influenced by the number of bedrooms compared to bathroom and floors
The age of houses varies based on their renovation status, providing a comprehensive overview.
The importance of recent renovation in boosting property sales.
House Age Distribution by Number of Bathrooms,Bedrooms and Floors
Age of House (in Years)
50
51
59
64
71
Value
1400
1300
1200
1100
1000
900
800
700
600
500
400
300
200
Measure Names
No of Bathrooms
No of Bedrooms
No of Floors
Drag to add text
Show title
Size
Story (1016 x 964)
Data Source
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Dashboard 2
Story 1
Dashboard 3
Story 2

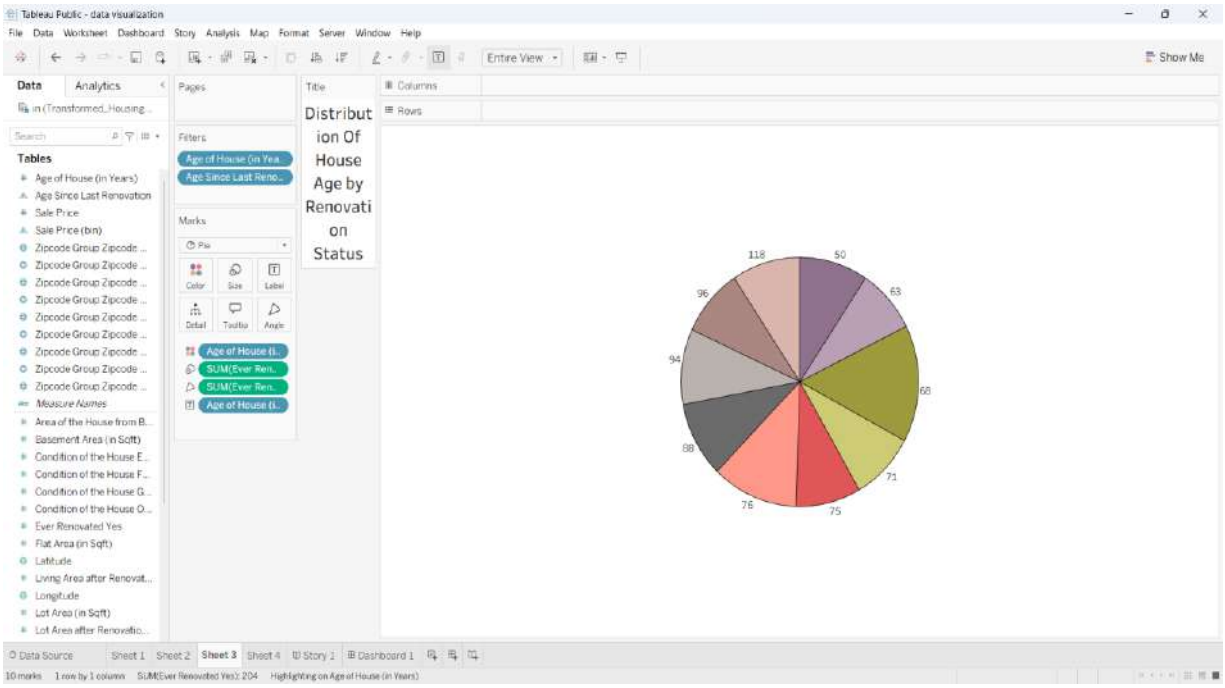


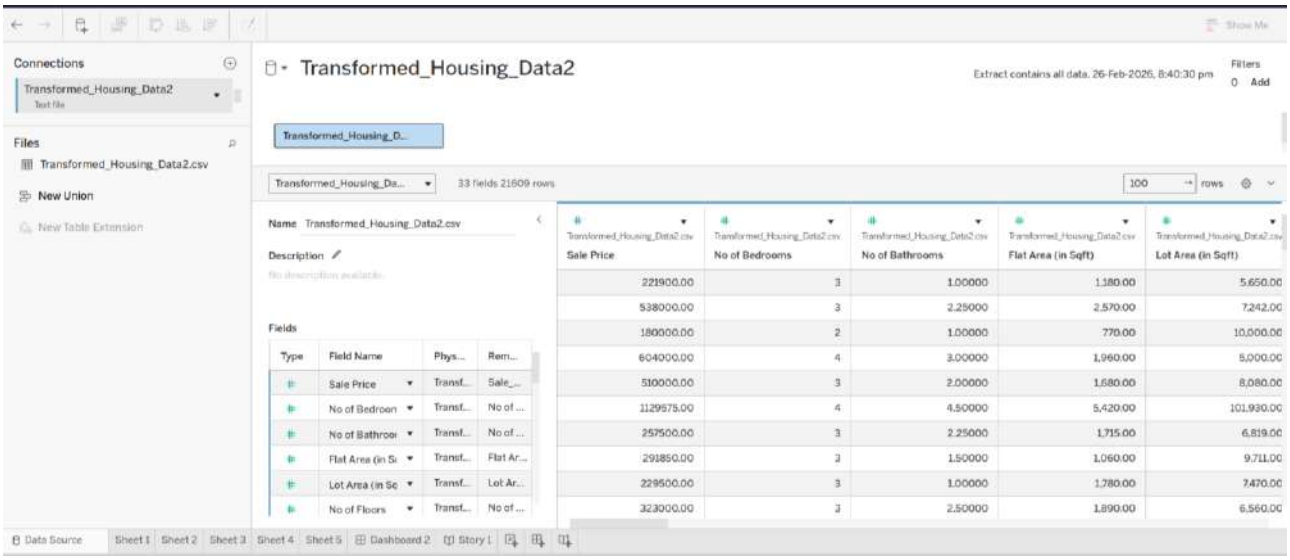
Tableau Public - data visualization
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Entire View
Show Me
Data
Analytics
Tables
Pages
Filters
Marks
Pie
Color
Size
Label
Detail
Tooltip
Angle
Title
Distribution Of House Age by Renovation Status
Columns
Rows
118
50
63
68
71
75
76
88
94
96
Data Source
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Story 1
Dashboard 1

5) Performance Testing

Performance testing involves assessing the volume of data rendered from the database, the impact of data filters on system responsiveness, and the complexity introduced by the number of visualizations. Optimizing these factors ensures the dashboard operates efficiently, providing timely and reliable insights.

❖ Amount of Data Loaded

"Amount of Data Loaded" refers to the quantity or volume of data that has been imported, retrieved, or loaded into a system, software application, database, or any other data storage or processing environment. It's a measure of how much data has been successfully processed and made available for analysis, manipulation, or use within the system.



❖ Utilization of Filters

"Utilization of Filters" refers to the application or use of filters within a system, software application, or data processing pipeline to selectively extract, manipulate, or analyse data based on specified criteria or conditions.

1) Selected "Sales Price" as a Top N Filters.

a) Drage and drop "Sales Price" into Filters Filed.
b) Right click on "Sales Price" and click on **Edit Filter**.
c) Then Click on **Top/Bottom**.

Filter [Sale Price]

General

List

Custom value list

Search or add text

Add

Search

Clear List

Exclude selected values

Include all values when empty

Wildcard

Condition

Top/Bottom

Summary

Selection: Selected all values
Wildcard: All
Condition: None
Limit: Top 25 by SUM([Age of House (in Years)])

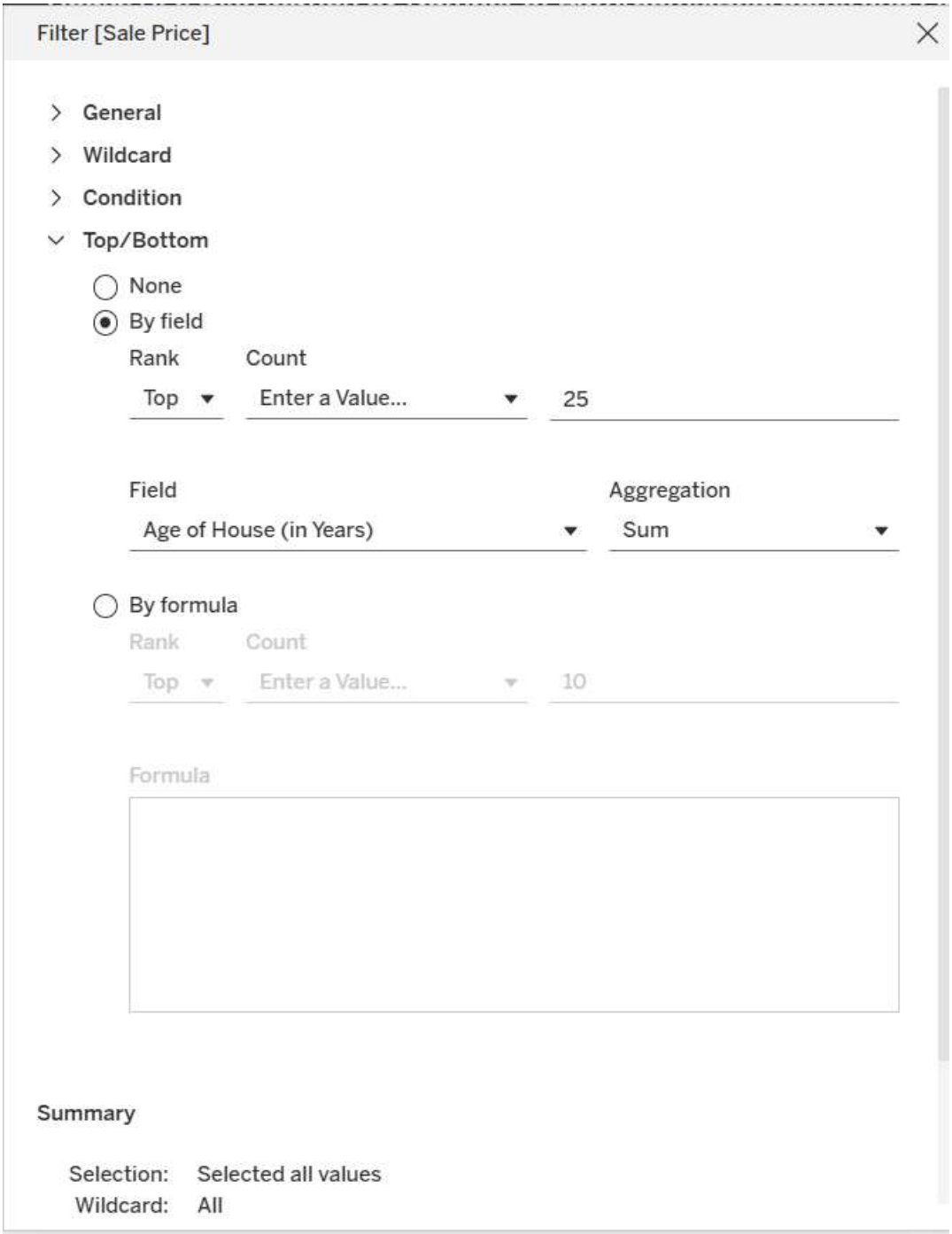
Reset

Apply

Cancel

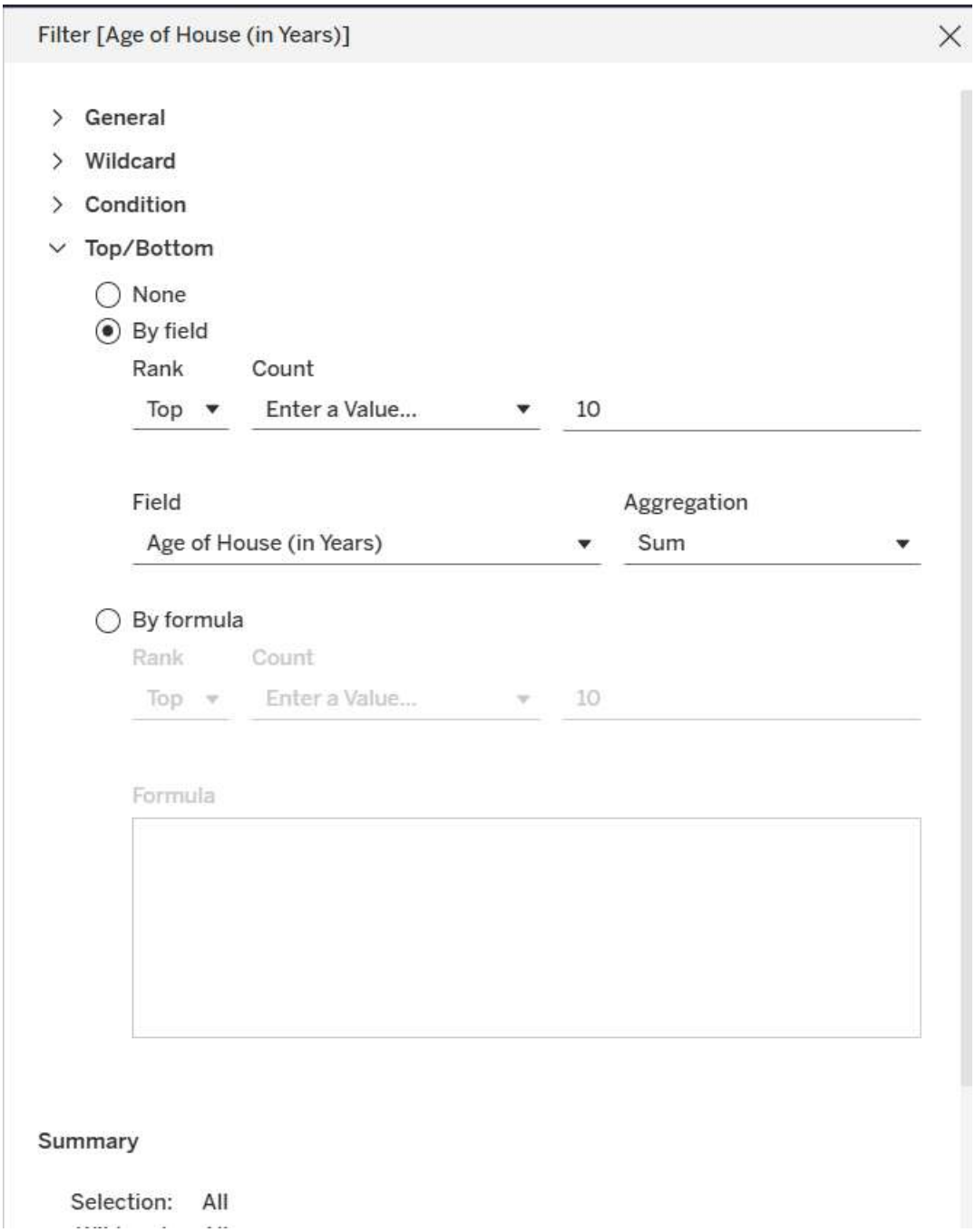
OK

d) Select **By Field** and add Top 25 records and click OK.



2) **Select "Age of House (in Year)" as a Top N records.**
   a) Drage and drop "Age of House (in Year)" into Filters Filed.
   b) Right click on "Sales Price" and click on **Edit Filter**.
   c) Then Click on **Top/Bottom**.

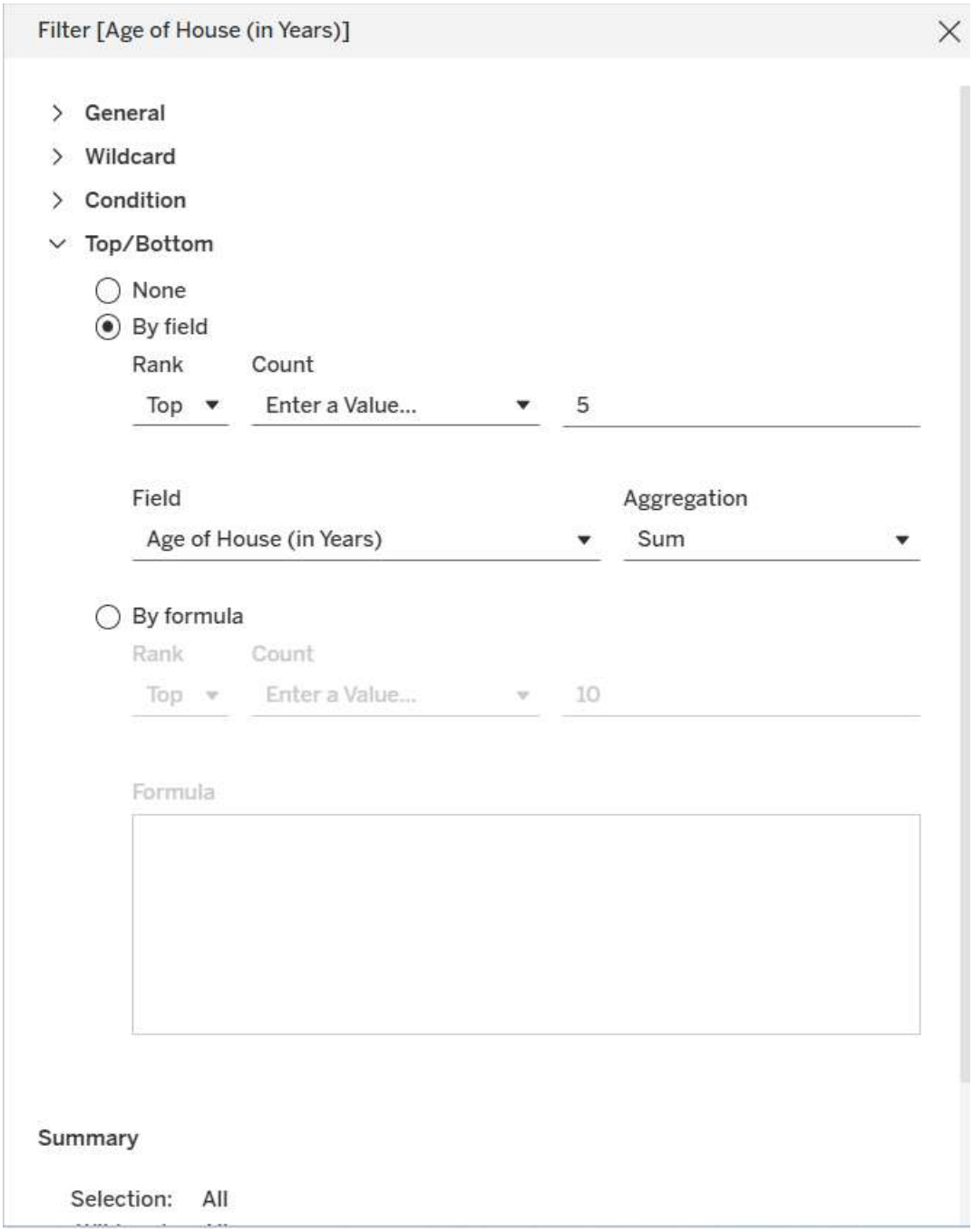
d) Select **By Field** and add Top 10 records and click OK.



3) **House Age Distribution by Top N Filters.**

a) Drage and drop “Age of House (in Year)” into Filters Filed.
b) Right click on “Sales Price” and click on **Edit Filter**.
c) Then Click on **Top/Bottom**.

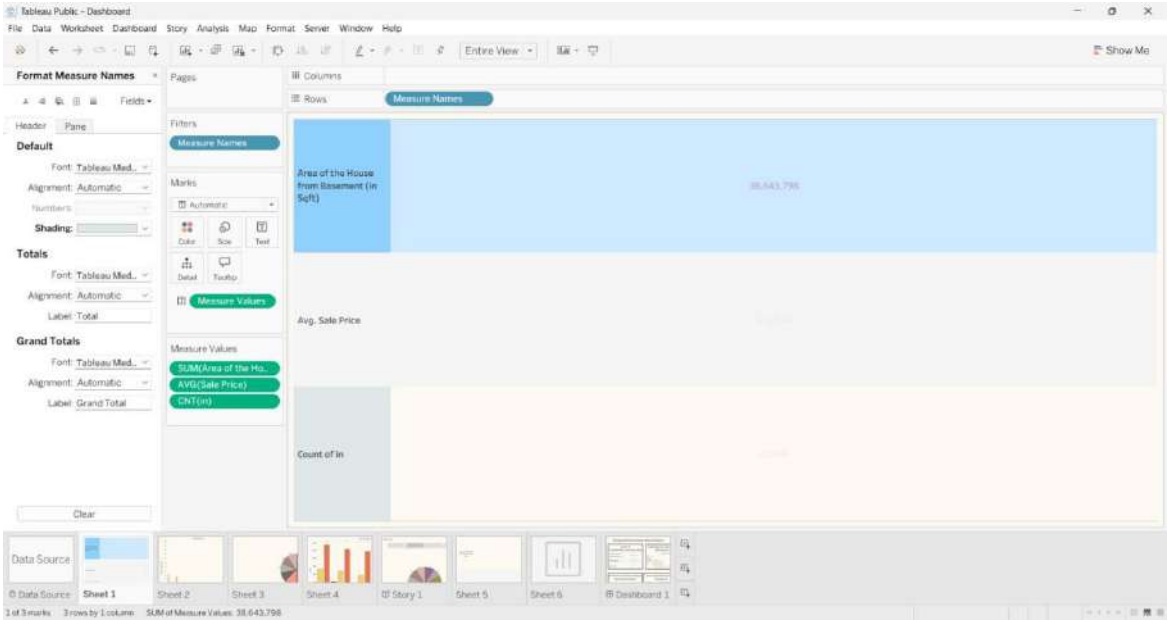
d) Select **By Field** and add Top 5 records and click OK.



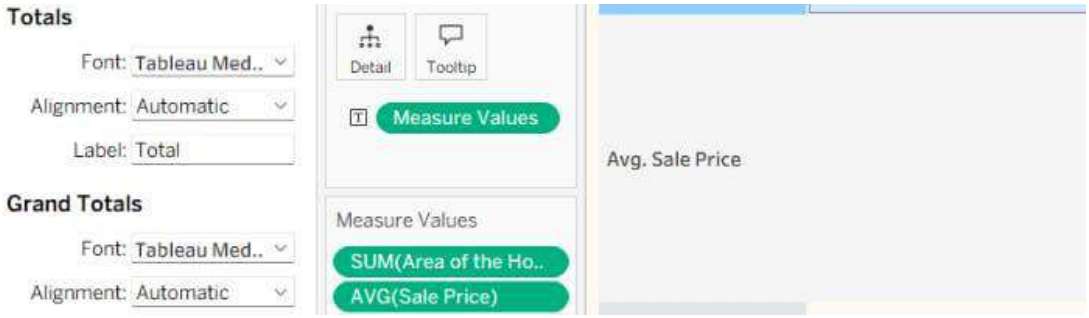
❖ No of Visualizations/ Graphs

1) Count of Transformed Housing Data

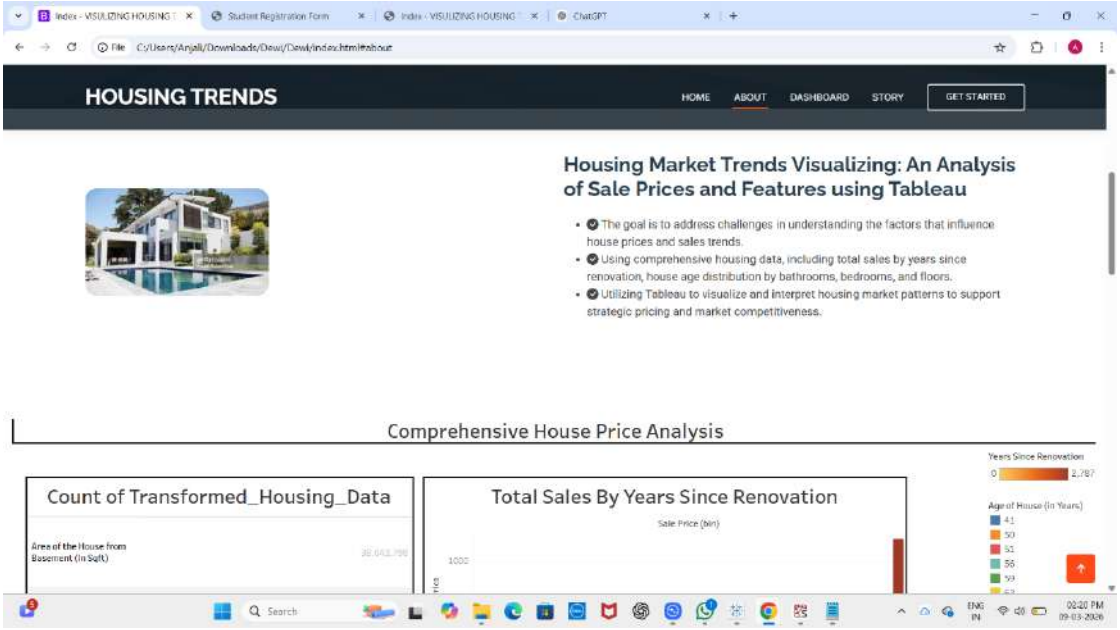
2) Average Sale Prices



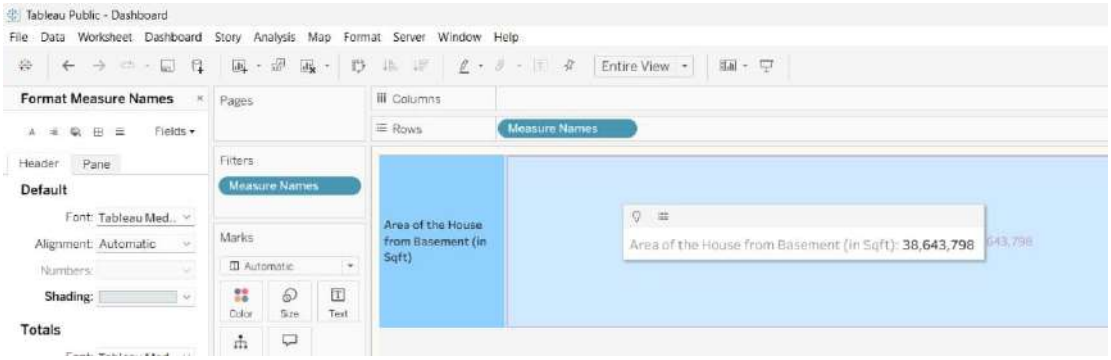
3) Area of House from Basement (in Sqft)



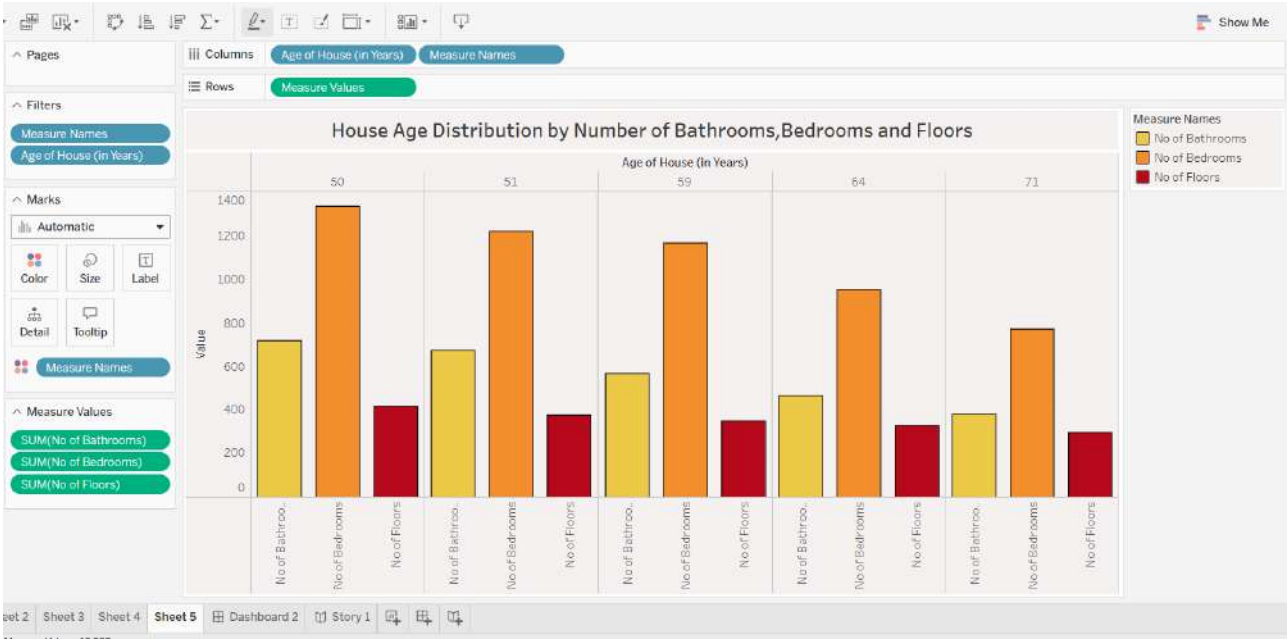
4)Total sales by years by renovation



5)Distribution of House Age by Renovation Status



6)House Age Distribution by Number of Bathrooms, Bedrooms, and Floors



4) Web integration

Web integration of a Tableau involves embedding interactive visualizations into a website or web application. This allows users to explore data insights directly within a web interface, enhancing accessibility and engagement. It supports real-time updates, user filtering, and seamless navigation for a dynamic data storytelling experience.

❖ Publishing

a) **Sign in to Tableau Public**

In Tableau Desktop, go to **File** > **Save to Tableau Public**.
If you're not already signed in, a login window will appear.
Enter your **Tableau Public credentials** or sign up if you don't have an account.

b) **Save and Publish**

- After logging in, you'll be prompted to **name your workbook**.
- Click **Save** – Tableau will upload the workbook to your Tableau Public profile.

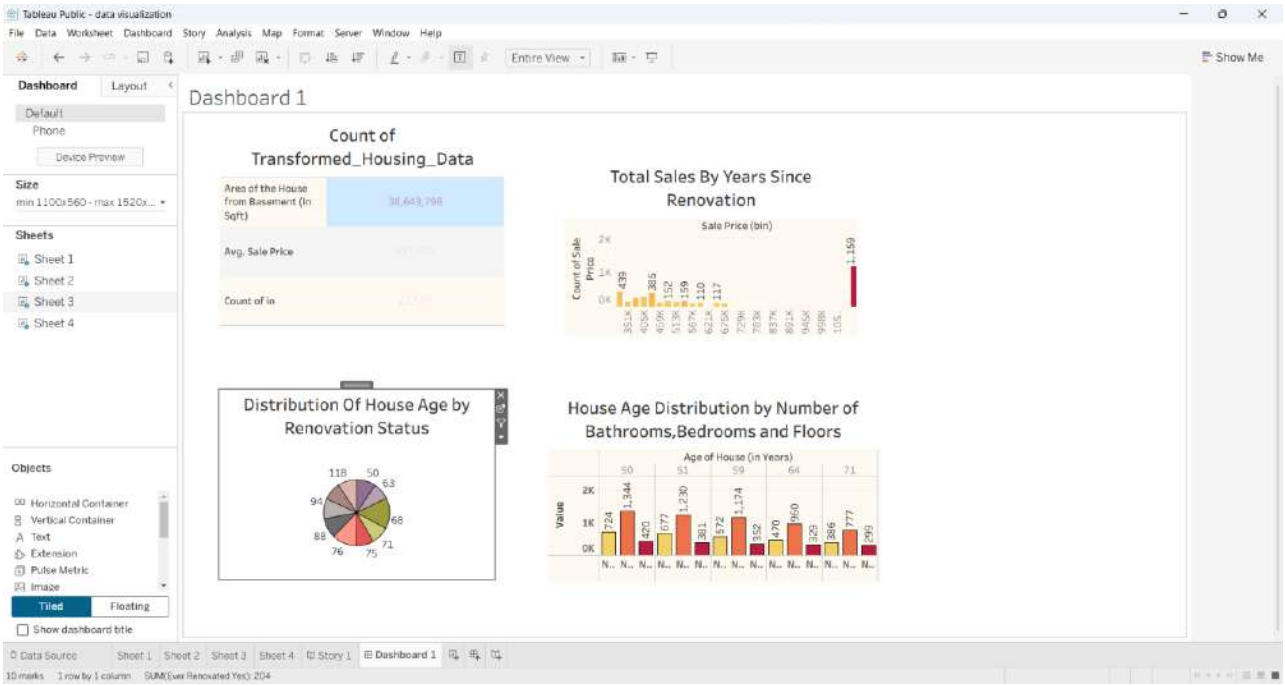


Tableau Public - data visualization
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Entire View
Show Me
Dashboard
Layout
Default
Phone
Device Preview
Size
Sheets
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Objects
Horizontal Container
Vertical Container
Text
Extension
Pulse Metric
Image
Tiled
Floating
Show dashboard title
Dashboard 1
Count of Transformed_Housing_Data
Area of the House from Basement (In Sqft)
Avg. Sale Price
Count of in
Total Sales By Years Since Renovation
Sale Price (bin)
Count of Sale Price
Distribution Of House Age by Renovation Status
House Age Distribution by Number of Bathrooms,Bedrooms and Floors
Age of House (in Years)
Value
Data Source
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Story 1
Dashboard 1
10 marks
1 row by 1 column
SUM(Ever Renovated Yes): 204

public.tableau.com/app/profile/shreya.pawar5452/viz/HouseAgeDistributionbyNumberofBathroomsBedroomsandFloors/Story1
tableau public
Create
Learn
Search for vizzes, authors, and hashtags
Register for Tableau Conference 2026 and save $300 → Last Chance Pricing
House Age Distribution by Number of Bathrooms,Bedrooms and Floors by Shreya Pawar
House Age Distribution by Number of Bathrooms,Bedrooms and Floors
Age of House (in Years)
Measure Names
No of Bathrooms
No of Bedrooms
No of Floors
Value

❖ Dashboard and Story embed with UI with Flask.

In this activity, you will create a simple web application using Flask and embed a Tableau Dashboard or Story into the UI. This allows users to view and interact with Tableau visualizations directly through a custom web interface.

1) **temp.py**

2) **Adding Dashboard and Story on HTML page.**
   a) Open Tableau Public profile and open Dashboard/Story.
   b) Click on share option and copy the embed code.
   c) Paste in HTML page
   d) Run the temp.py and copy the port number and paste it on Browser.